

COLLABORATIVE NOMA-ISAC VIA MULTI-AGENT PPO

BEYOND INDEPENDENT
LEARNING IN UAV NETWORKS

PRESENTATION BY
RUDRANSH VERMA
B.TECH CSE DEPT.

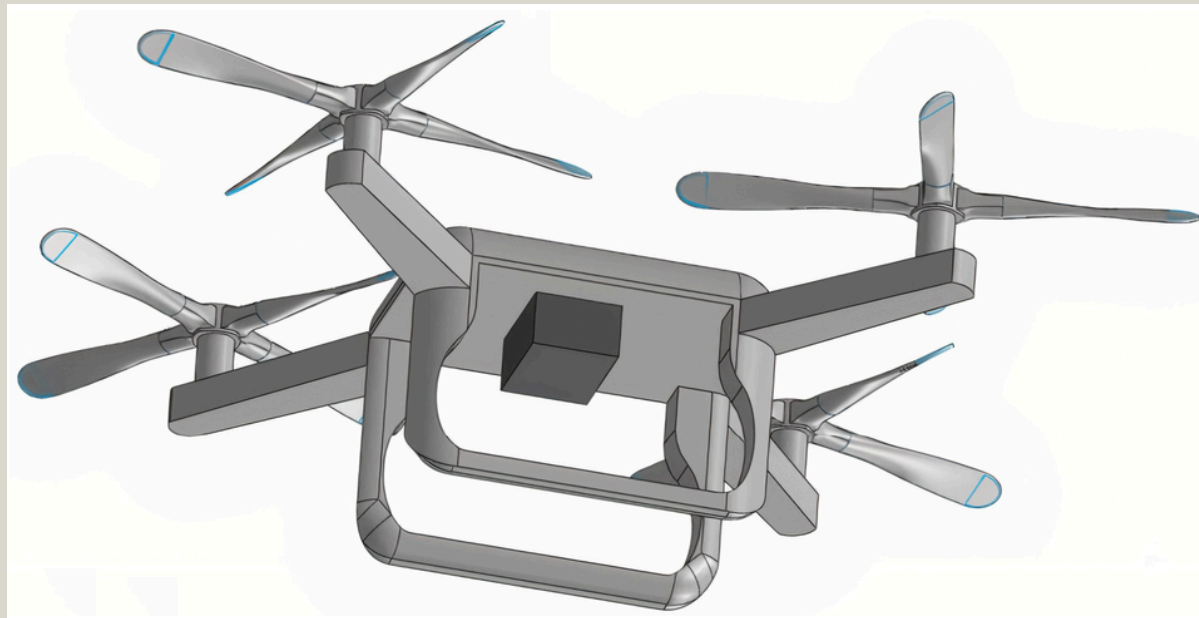
CONTENT PILLARS

- ➔ **Introduction: ISAC and UAVs**
- ➔ **Motivation towards ISAC-NOMA via UAVs**
- ➔ **Problem and recent works on solving it**
- ➔ **Limitations and other Literature Reviews**
- ➔ **Our proposed architecture**
- ➔ **How the new approach solves the limitations**

ISAC and UAVs

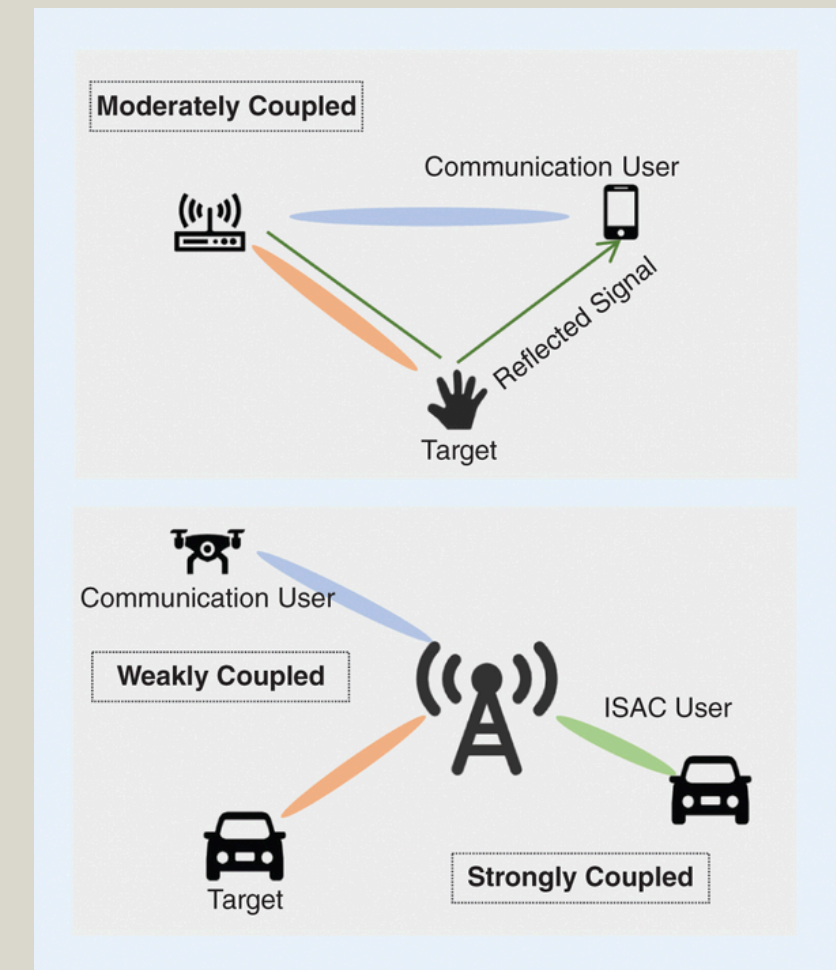
Integrated Sensing and Communication (ISAC)

A foundational 6G pillar that merges **Radar Sensing** and **Wireless Communication** into a single hardware platform and unified waveform, improving spectral efficiency and hardware utilization by allowing signals to “see” and “talk” simultaneously.



A UAV, commonly known as a **drone**, is an aircraft operated without a human pilot on board. It is controlled remotely by a human operator or autonomously via onboard sensors, computers, and GPS. They have many advantages when used for wireless communication instead of static base stations:

- a) UAVs can adjust their 3D coordinates to optimize quality
- b) Their altitude minimizes “shadowing” from buildings
- c) Rapidly deployable for disaster relief/high-density events



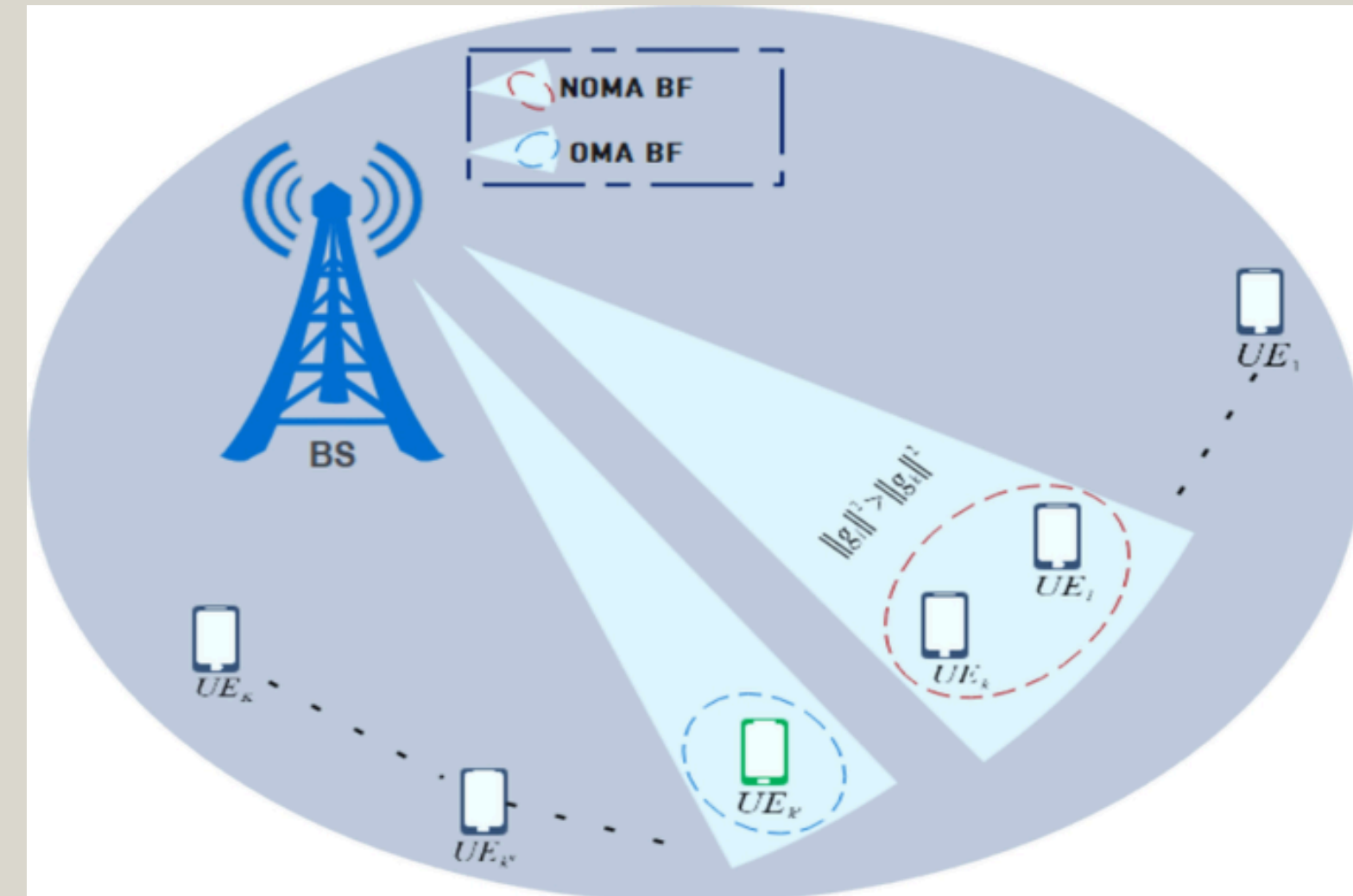
Unmanned Aerial Vehicles (UAVs)

Introduction

UAV-assisted networks have become indispensable for emergency communications where terrestrial base stations are destroyed or unavailable.

Non-Orthogonal Multiple Access (NOMA) uses power-domain multiplexing to overcome the spectral inefficiency of traditional Orthogonal Multiple Access (OMA).

We deploy a swarm of UAVs, each acting as an autonomous agent that must collaborate to provide seamless coverage.



Motivation

The Crisis of Density

In large-scale events or disasters, OMA is unsustainable for dynamic populations; NOMA can achieve over 128% higher spectral efficiency in dense scenarios.

Hardware Synergy

Integrating NOMA with ISAC resolves two major trade-offs: it supports high-density connectivity while enabling unified sensing and communication operations on shared resources.

The Necessity of Autonomy

Traditional optimization fails in stochastic wireless environments. We need a system that "learns" the environment through Multi-Agent Deep Reinforcement Learning to optimize trajectories and power in real-time.

The Ultimate Goal

To minimize downlink Transmission Delay (TD) while maximizing Conditional Mutual Information (MI) to ensure that information flows fast and environmental sensing is sharp.

ISAC and UAVs

Now the question arises...

- A) Where should the drone fly?
- B) Which users should share the same beam?
- C) How much energy goes to each user's NOMA layer?

These three variables are highly coupled and Non-Convex, hence the need for Reinforcement Learning

The Dueling DQN Approach

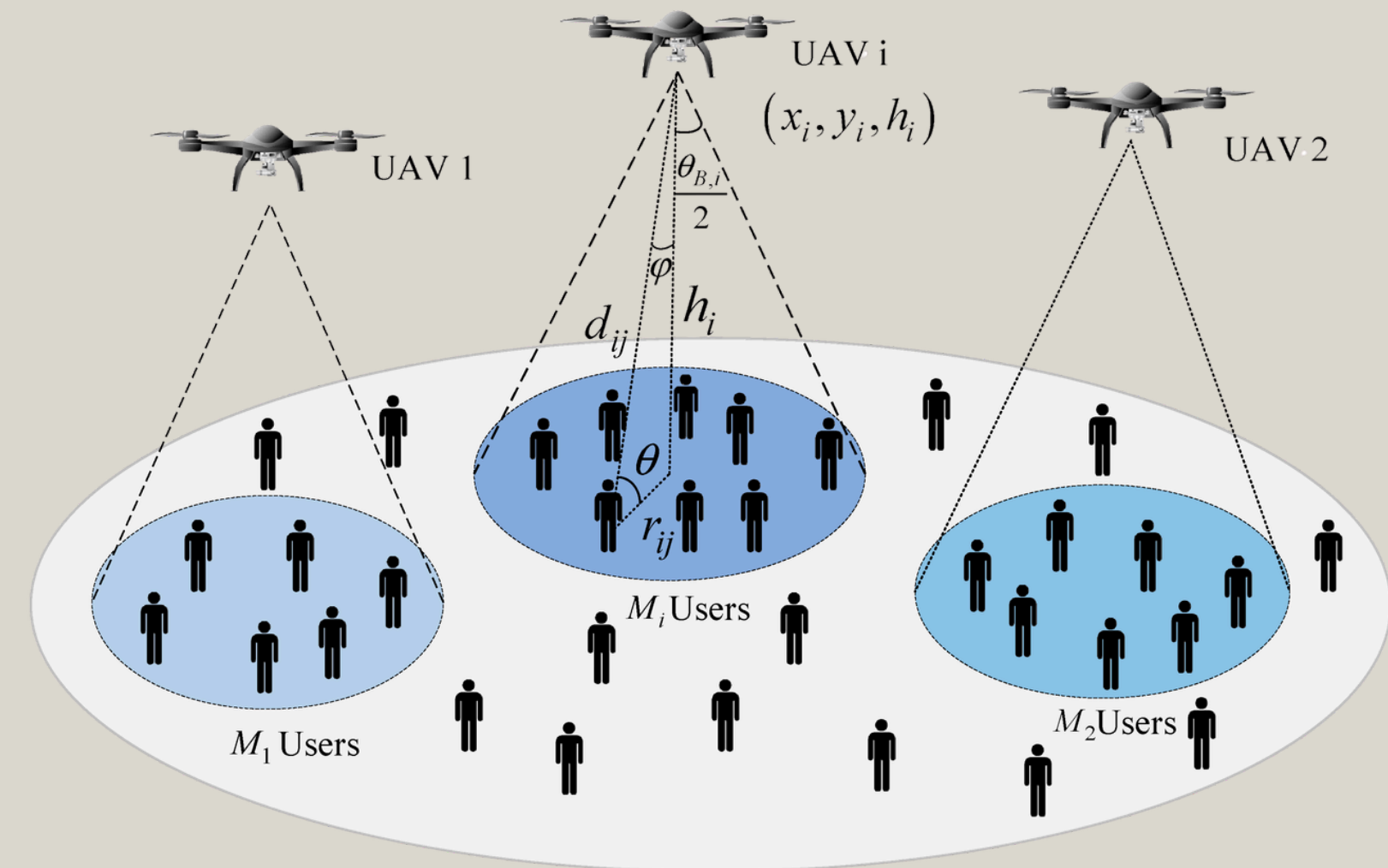
The Study: “NOMA-ISAC Empowered UAV Networks” (Zhou et al., 2025)

Clustering via DWAHC

Density-Weighted Agglomerative Hierarchical Clustering: Unlike simple K-means which looks for only distance, this algorithm groups users based on their spatial density and proximity.

Dynamic Load Balancing: Adjusts clusters to ensure no single UAV is overloaded preventing bottlenecks in Transmission Delay (TD).

We use these two techniques for periodic cluster updates in every M-slot intervals to ensure adaptability.



The Dueling DQN Approach

The Study: “NOMA-ISAC Empowered UAV Networks” (Zhou et al., 2025)

Optimization Problem of UAV

We need to optimize the flight trajectory $\mathbf{L}$ and transmit power $\mathbf{P}$ for each user group, minimizing average total TD ($\tau(t)$) and maximize average total Conditional Mutual Information ($I(t)$).

Mathematically, this can be represented as:

$$\max_{\{\mathbf{L}, \mathbf{P}\}} \sum_{t=0}^T \alpha \frac{1}{\tau(t) \tau_{\text{scale}}} + (1 - \alpha) \frac{I(t)}{I_{\text{scale}}}$$

Subject to:

$$\begin{aligned} h_{\min} &\leq h_u(t) \leq h_{\max}, \quad \forall u, t & \sum_{k=1}^{K_u} P_u^k(t) &\leq P_{\max}, \quad \forall u, t \\ x_{\min} &\leq x_u(t) \leq x_{\max}, \quad \forall u, t & G_u^k(t) &\geq G_u^m(t), \quad \forall u, k, t \\ y_{\min} &\leq y_u(t) \leq y_{\max}, \quad \forall u, t & R_u^k(t) &\geq R_{\text{th}}, \quad \forall u, k, t \\ L_u(t) &= L_s(t), \quad \forall u, s, t & I_u^k(t) &\geq I_{\text{th}}, \quad \forall u, k, t \end{aligned}$$

The Dueling DQN Approach

The Study: “NOMA-ISAC Empowered UAV Networks” (Zhou et al., 2025)

Optimization Problem of UAV

Dueling DQN architecture: This splits the network into two streams: one estimates the state value $V(s)$, and another for action advantage $A(s,a)$. This helps learn which states are more valuable without needing to evaluate every single action in that state.

Discrete State Space: The input state contains 3D coordinates of all UAVs and channel gains between all UAVs and users of UAV u .

$$S = \{L_u(t), L_s(t), g_k^u(t), g_k^s(t)\}, \forall u, s, k, t$$

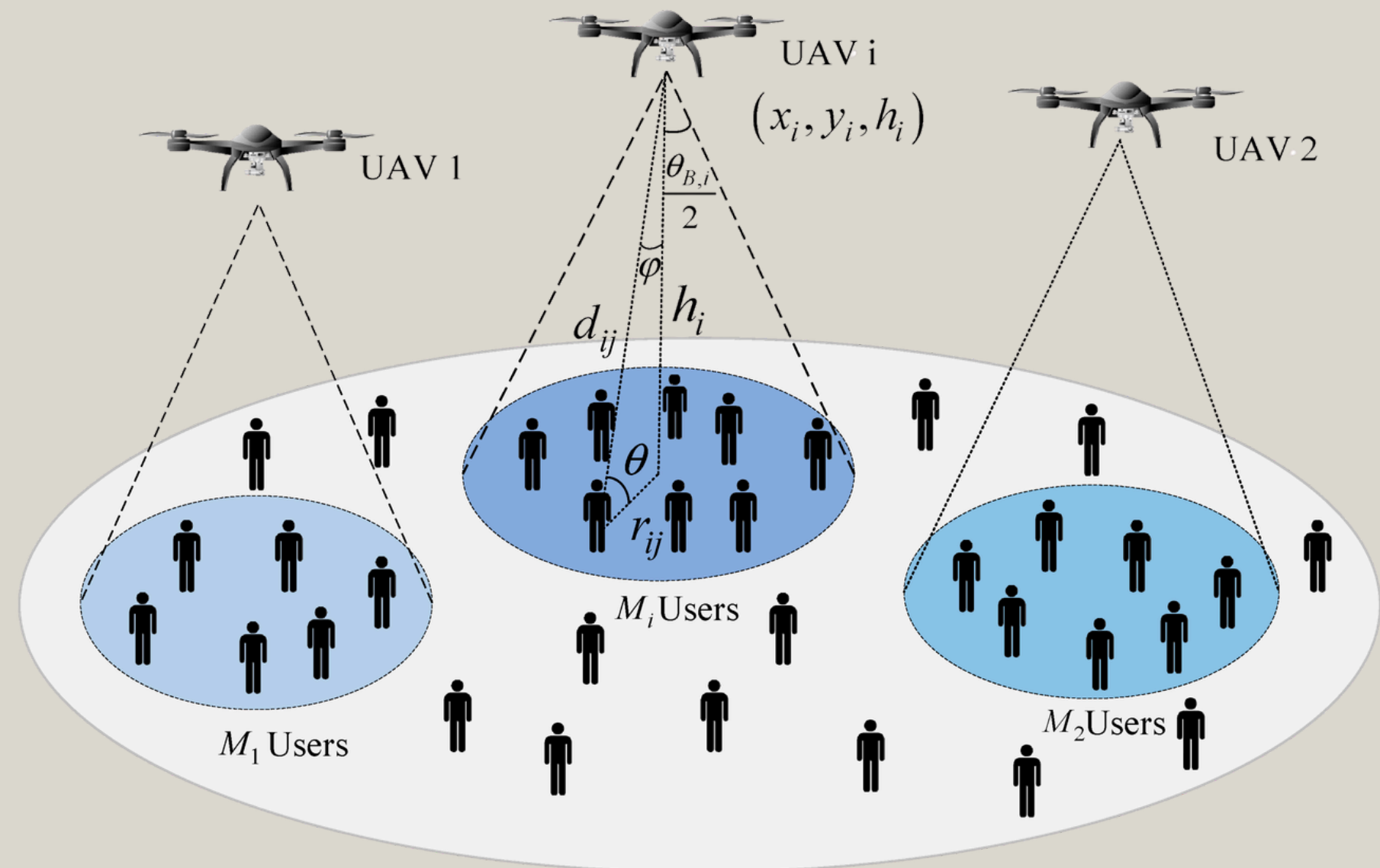
Discrete Action Space: 7 fixed maneuvers of UAV (forward, back, left, right, up, down, hover) and 3 predefined user power levels (P_1, P_2, P_3).

The Dueling DQN Approach

The Study: “NOMA-ISAC Empowered UAV Networks” (Zhou et al., 2025)

Optimization Problem of UAV

Each UAV acts as an **Independent Agent**, but the reward function is set in such a way that it focuses on the whole system's performance.



The Dueling DQN Approach

The Study: “NOMA-ISAC Empowered UAV Networks” (Zhou et al., 2025)

Optimization Problem of UAV

Estimating Q-Values: For the Dueling architecture, the Q-value of taking action A_u under state S_u is composed of the state value function $V(S_u; \theta_v)$ and the advantage function $A(S_u, A_u; \theta_a)$, denoted as $Q(S_u, A_u; \theta_v, \theta_a)$.

Action Policy Selection: Throughout the entire training process, the ϵ -greedy action policy is used to maintain a balance between exploration and exploitation

Reward Function Design: The proposed reward function for **each agent** incorporates an exponential penalty term (the denominator), halving the reward upon Quality of Service (QoS) threshold violations.

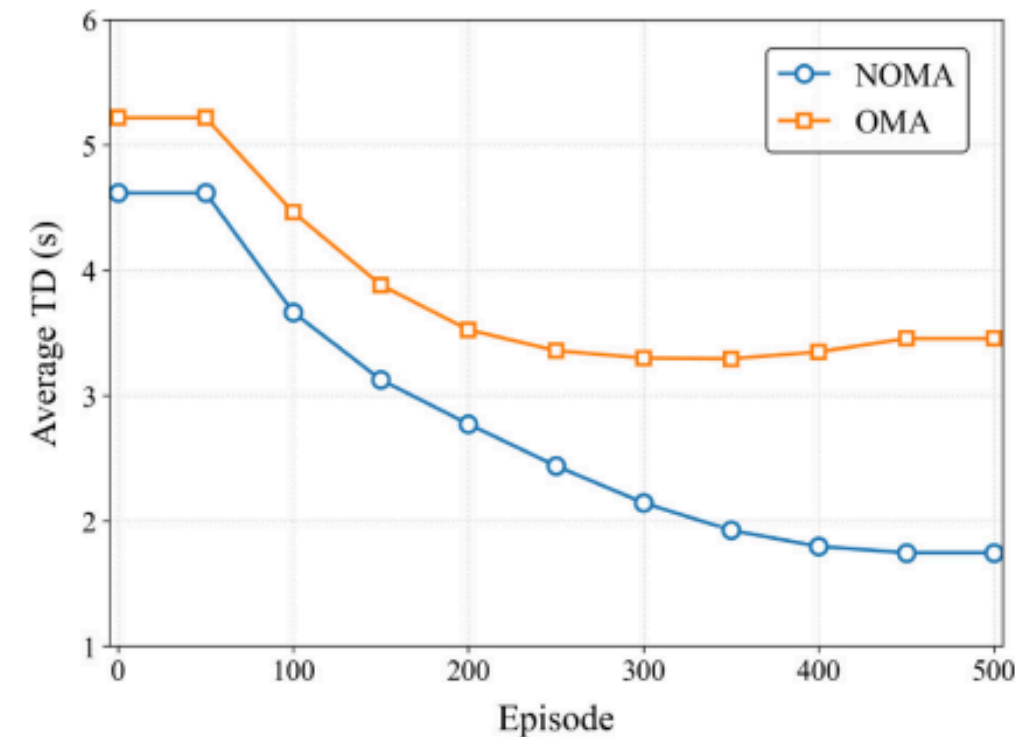
$$r(t) = \frac{\alpha \frac{1}{\tau(t) \tau_{\text{scale}}} + (1 - \alpha) \frac{I(t)}{I_{\text{scale}}}}{2^{\max(\mathbb{I}(R_k^u(t) < R_{\text{th}}), \mathbb{I}(I_k^u(t) < I_{\text{th}}))}}$$

The Dueling DQN Approach

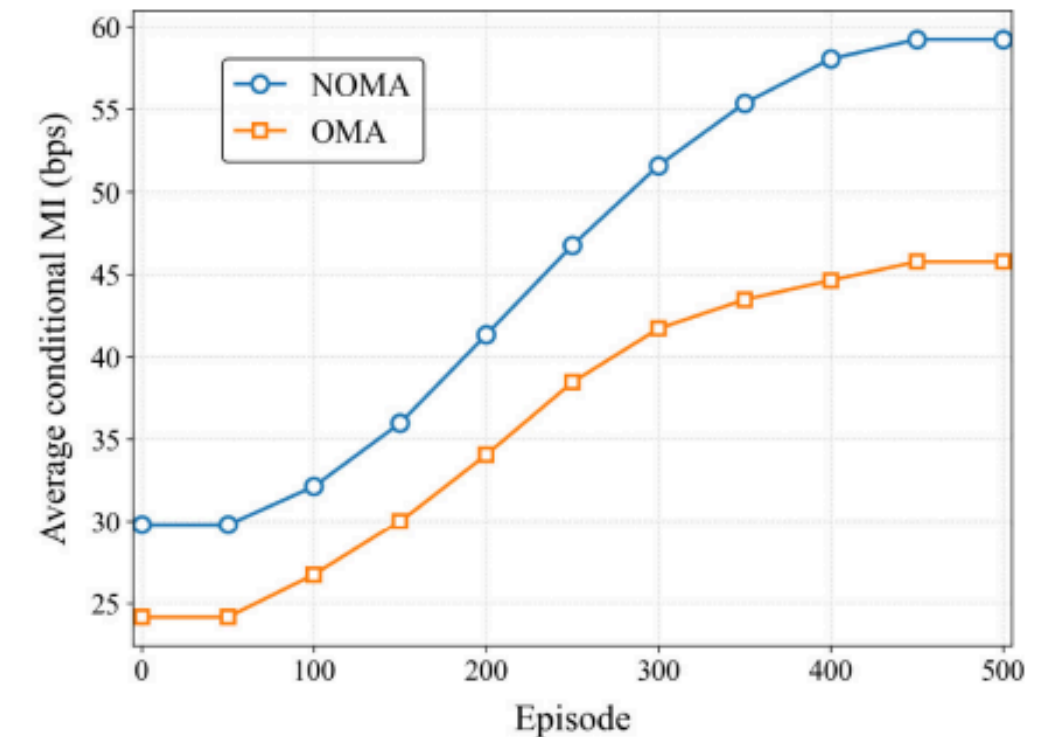
The Study: “NOMA-ISAC Empowered UAV Networks” (Zhou et al., 2025)

Reported Benchmarks

- Reduced Transmission Delay by **14%-22%** compared to OMA.
- Maintained Sensing Mutual Information (I) above the required threshold (I_{th})



(a) Average TD.



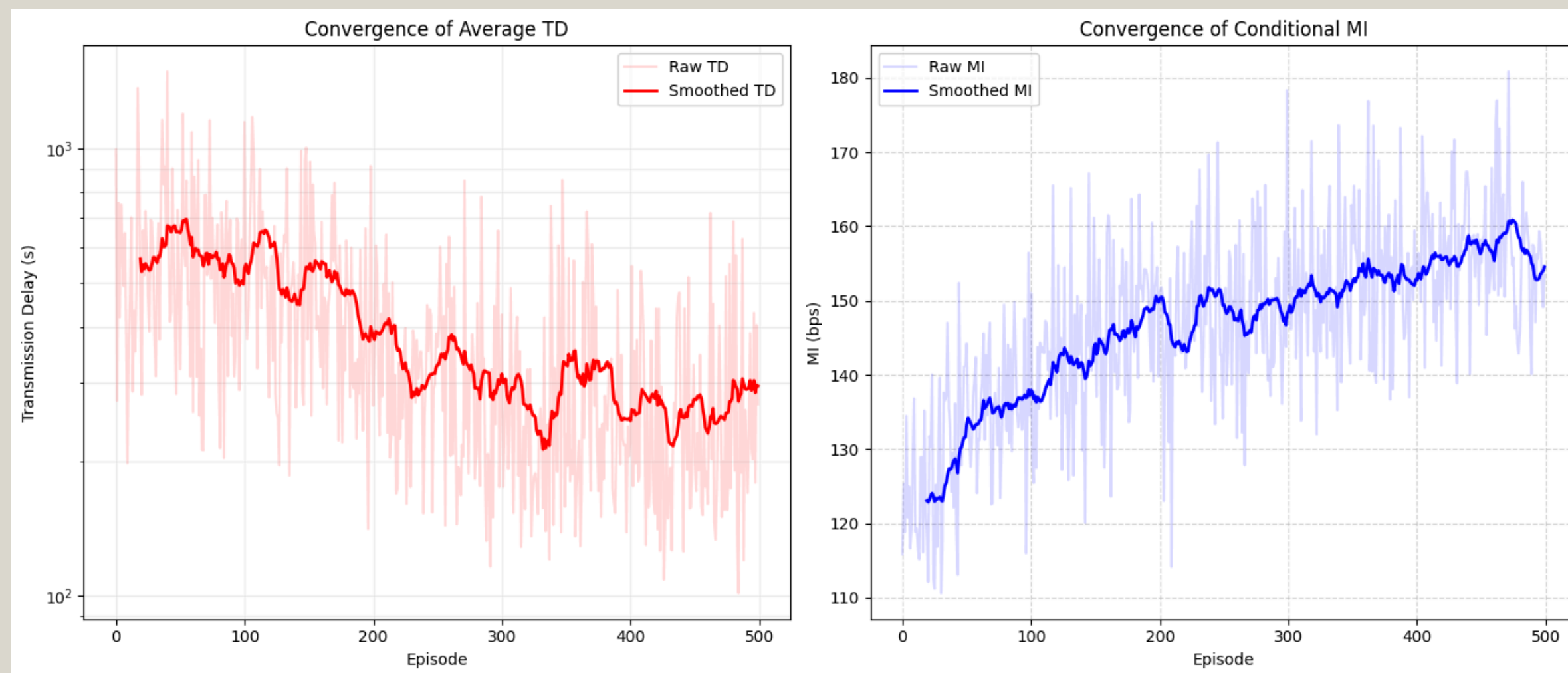
(b) Average conditional MI.

We take this model as the baseline model for our improved model.

The Dueling DQN Approach

The Study: “NOMA-ISAC Empowered UAV Networks” (Zhou et al., 2025)

Simulating the results locally



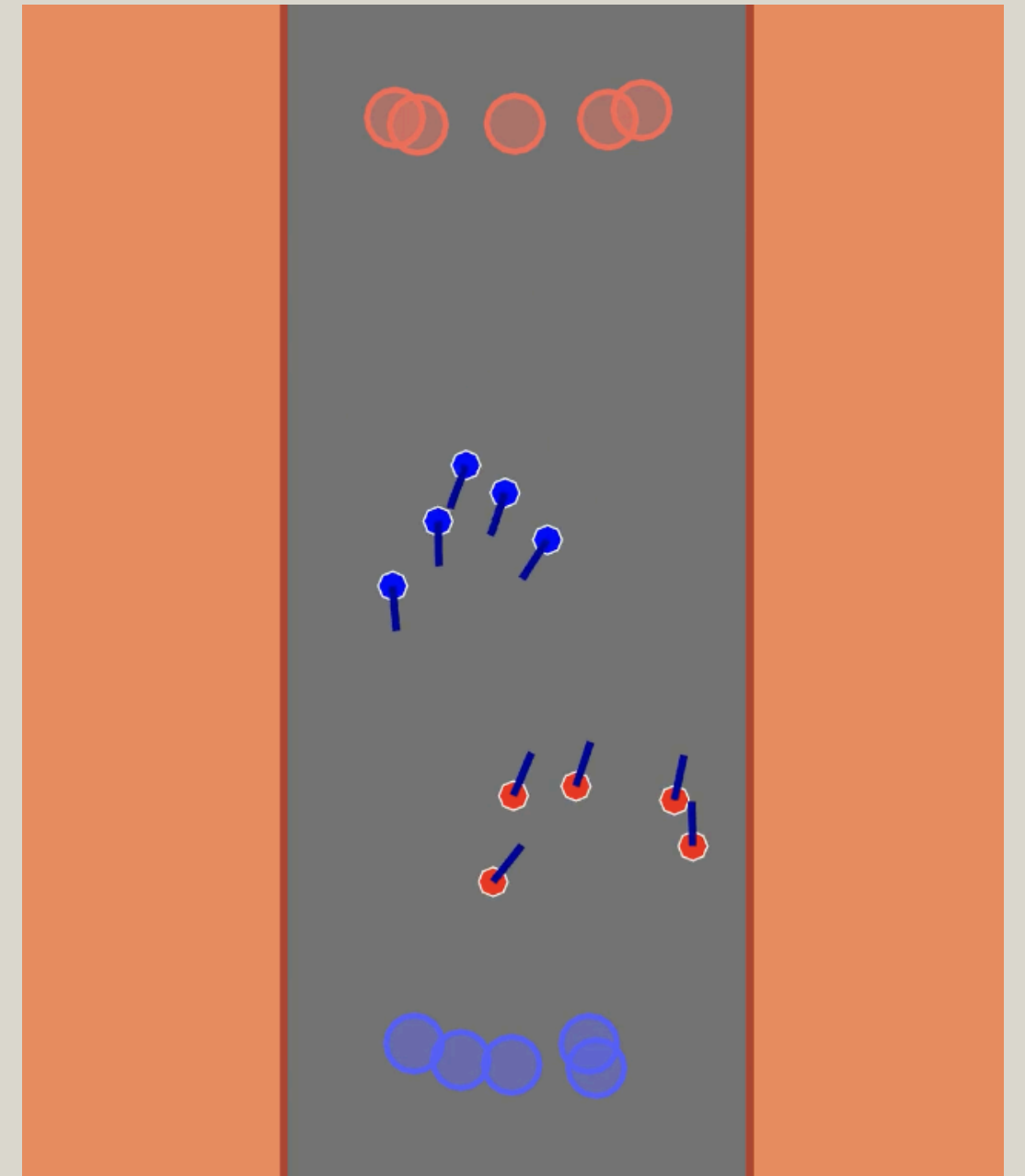
- These are the results of Transmission Delays and Conditional Mutual Information trend with increasing episodes, using the discussed approach.

Why Independent DDQN Learning fails

Non-Stationarity:

Each UAV is an Independent learner so when UAV A updates its policy and moves, the environment for UAV B changes. To UAV B, the state transitions and rewards seem to be constantly shifting. This leads to a lack of convergence guarantees. The agents are basically trying to hit a moving target.

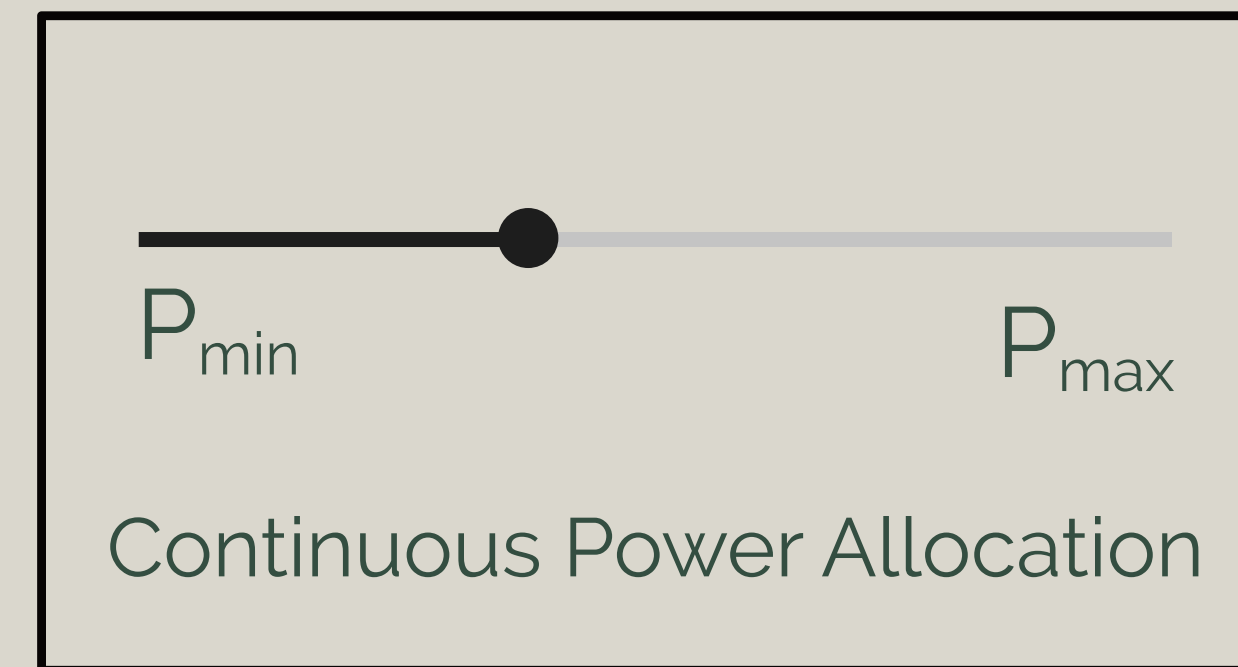
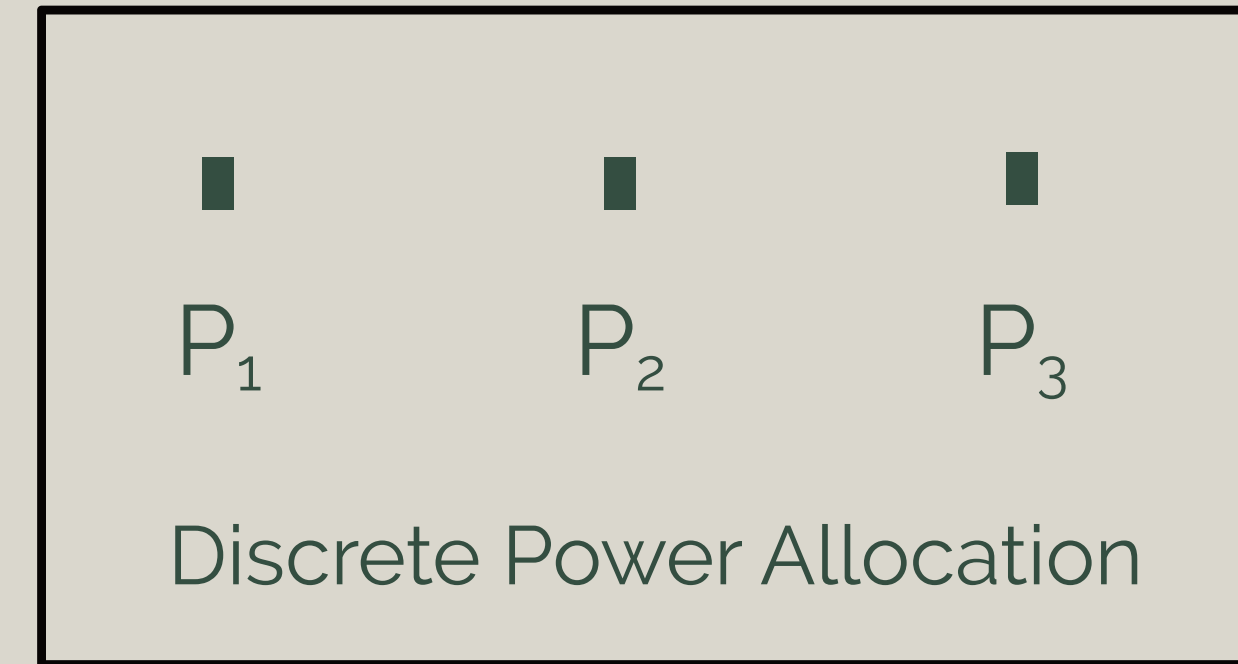
Essentially, since these agents do not interact with each other, they will single-handedly optimize the global reward which leads to non-cooperation.



Why Independent DDQN Learning fails

Discrete Action Space:

The Baseline uses 7 maneuvers and 3 power levels. The Real-World 6G requires high-fidelity control. Discrete steps lead to “jerky” trajectories and “near optimal” power allocations that misses the fine-tuned balance needed for perfect NOMA Successive Interference Cancellation (SIC).



Why Independent DDQN Learning fails

Unstable/Bad Episodic Learning

Q-learning (DDQN) is notoriously unstable. One bad experience can cause the UAV to "panic" and drastically change its strategy, which then confuses all the other UAVs.

Consider a situation where one UAV suddenly encounters a massive burst of noise, the DDQN might "flip out" and max out its power. This leads from a bad action to a worse state consequently to a worse action, thus making it an even worse episode than it should have been,

Proposed Architecture: Collaborative MAPPO (Multi-Agent Proximal Policy Optimization)

Centralized Training, Decentralized Execution (CTDE)

UAVs learn to cooperate in a simulation environment where they have access to “Global Truth”, but they operate independently in the real world using “Local Senses”.

1 Centralized Critic (The Global Brain):

It is used only during training. It takes a Global State (S) comprising positions of all UAVs and channel gains of all users. It will evaluate the “value” of joint actions, ensuring that one UAV's move doesn't negatively impact another's QoS (Quality of Service).

U Local Actors (The Onboard Controllers):

It is used during both training and execution. It takes in Location Observations (O_u) comprising of UAV u 's own position, assigned users, and immediate neighbors. It will be used to output real-time continuous commands for 3D velocity and NOMA power allocation.

Proposed Architecture: Collaborative MAPPO (Multi-Agent Proximal Policy Optimization)

Centralized Training

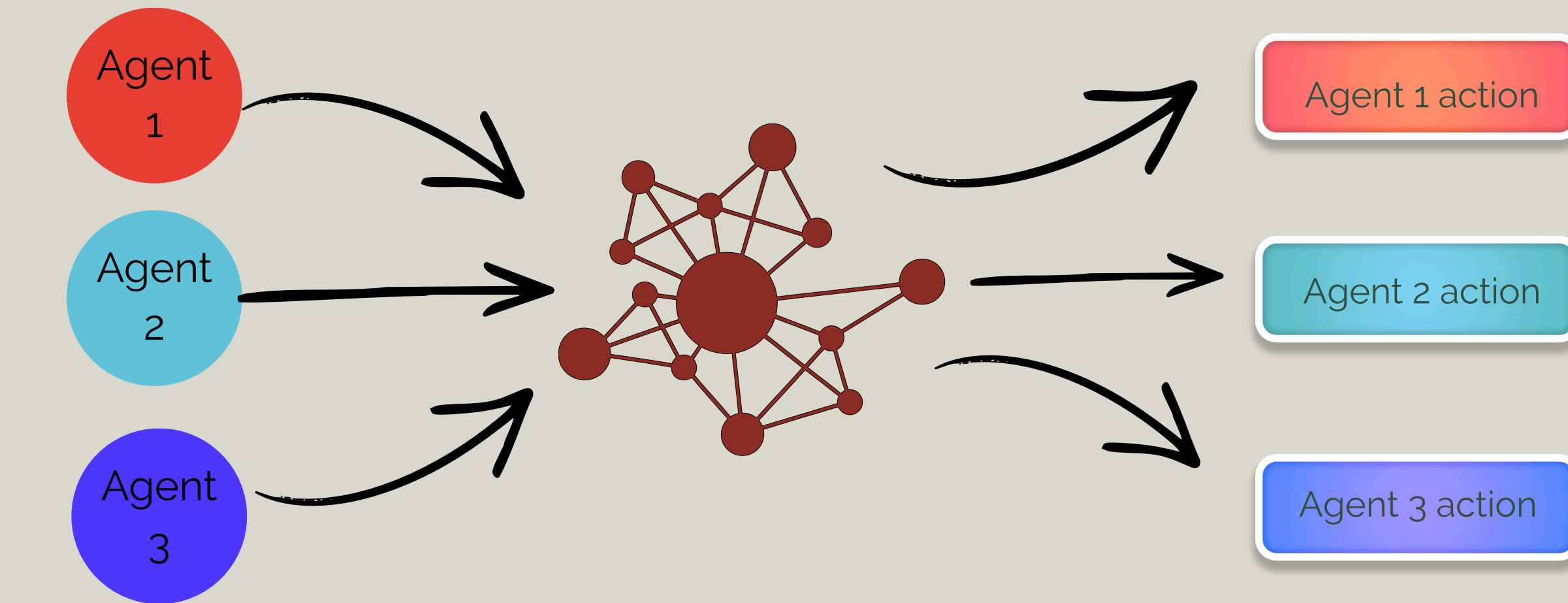
During training, the critic receives global information and learns how good global states are for the entire group.

The Actors receive local information but learn from the centralized critic.

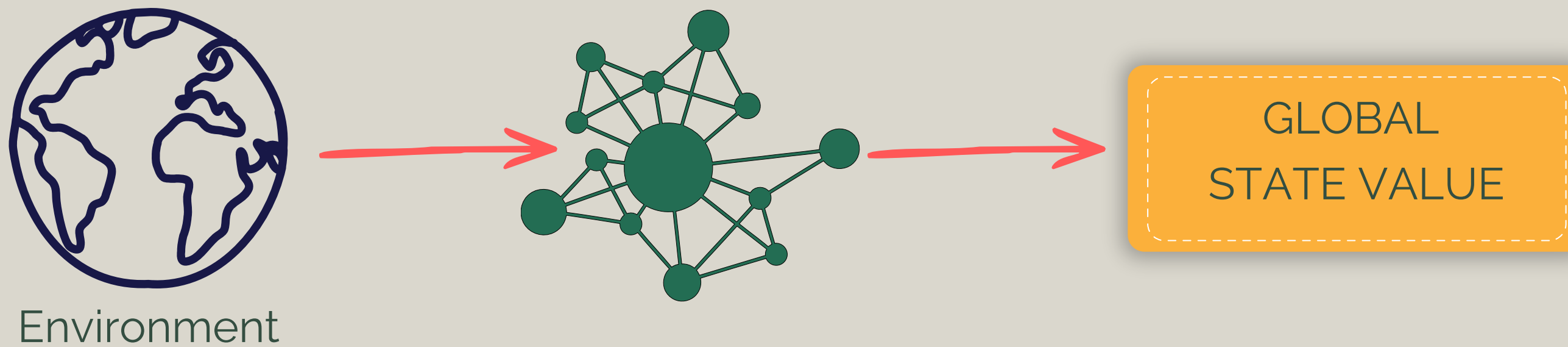
Decentralized Execution

During inference, the critics are no longer needed.

The Actors (agents) act according to their local spaces but still cooperate as a team.



Each agent independently generates an action with the same actor network



A single global value is returned to the agents representing how good the overall state is.

Proposed Architecture: Collaborative MAPPO (Multi-Agent Proximal Policy Optimization)

The Critic returns ONE value for all agents.
It holistically evaluates the state of the whole world.

Since the Actor networks try to maximize this advantage, they automatically learn cooperative behaviour that benefit the entire group.

Proposed Architecture: Collaborative MAPPO (Multi-Agent Proximal Policy Optimization)

This architecture solves all the three problems!

A) Solves Non-Stationarity: The Critic provides a stable target for learning, even when all UAVs are moving and changing the environment simultaneously.

B) Precision Control: Outputs smooth, continuous values ($P \in [0, P_{\max}]$) and velocity vectors, allowing for perfect SIC decoding that discrete steps might miss.

C) Clipped Stability: The "P" in PPO stands for Proximal. It has a built-in "safety belt" called Clipping. This mathematically prevents any UAV from changing its strategy too much in a single update.

Proposed Architecture: Collaborative MAPPO (Multi-Agent Proximal Policy Optimization)

Why this architecture Wins?

A) Global Objective Maximization: Unlike each independent DQNs maximizing global reward by their local action, MAPPO maximizes the Global Reward, $R_{\text{total}} = \sum_{u=1}^U r_u(t)$ where $r_u(t)$ is the local reward for the agent (UAV) u .

B) Scalability: MAPPO handles an increasing number of UAVs better than independent learners because the Centralized Critic accounts for the growing complexity of the inter-cluster interference map.

THANK YOU